

M16 Protected Review & Repair

Open only after genuine saved attempts

Standalone DE 2026 canonical rebuild - module-level QA candidate

DE16-01 Review / Repair - Production mindset

OPEN ONLY AFTER your independent attempt is saved.

What strong evidence shows

Strong evidence separates happy-path success from operational readiness: owner, blocking controls, monitoring, rollback/recovery and post-recovery validation.

Repair protocol

1. Identify the exact weak competency. 2. Close this Review. 3. Change the input/environment/failure condition. 4. Rebuild independently. 5. Preserve the original weak attempt. 6. Save new evidence and explain why it is stronger.

Do not repair by copying this paragraph into your answer. A changed task must still work.

DE16-02 Review / Repair - Testing strategy for data engineering

OPEN ONLY AFTER your independent attempt is saved.

What strong evidence shows

A strong strategy maps tests to risks. Unit tests cannot prove permissions or live integration; end-to-end tests are not a substitute for fast local logic tests.

Repair protocol

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DE16-03 Review / Repair - Automated quality gates

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What strong evidence shows

Blocking severity is policy-based, not a percentage. Duplicate business keys and impossible negative required values are blocking in this lab; anomaly volume changes need context and SLA rules.

Repair protocol

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DE16-04 Review / Repair - Continuous Integration fundamentals

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What strong evidence shows

CI validates the candidate change automatically before merge/deploy; production health still needs runtime monitoring and deployment evidence.

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DE16-05 Review / Repair - GitHub Actions for data projects

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What strong evidence shows

Start GITHUB_TOKEN permissions at least privilege. OIDC requires id-token: write to request the token, but cloud access still depends on the cloud trust policy and exchanged short-lived credential.

Repair protocol

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DE16-06 Review / Repair - Dev, test and prod environments

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What strong evidence shows

One codebase plus controlled config is preferable to three drifting copies. Secrets stay outside committed environment files.

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DE16-07 Review / Repair - Deployment and rollback

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What strong evidence shows

Rollback needs version identity, trigger, health validation and data-migration safety. Code rollback does not automatically reverse data mutations.

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DE16-08 Review / Repair - Secrets and credential safety

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What strong evidence shows

If a real secret leaked, rotate/revoke it and address repository history as needed. Secret scanning/push protection reduce risk but do not replace safe secret design.

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DE16-09 Review / Repair - RBAC and least privilege

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What strong evidence shows

Least privilege is action + resource + scope. Permission errors are diagnostic evidence, not justification for permanent admin access.

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DE16-10 Review / Repair - PII, encryption and masking

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What strong evidence shows

Encryption protects bytes from unauthorized reading in certain contexts; it does not authorize every user. Minimize and restrict access first, then select masking/encryption/tokenization appropriate to use.

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DE16-11 Review / Repair - Metadata, lineage and governance

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What strong evidence shows

Useful lineage records transformation relationships and downstream impact, not just a list of files. Metadata needs meaning, owner and operational context.

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DE16-12 Review / Repair - Observability for data pipelines

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What strong evidence shows

Observe freshness, volume, quality, duration/latency and status. An alert is useful only when an owner knows what action to take.

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DE16-13 Review / Repair - Incidents, runbooks and root cause

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What strong evidence shows

Do not call the first error line the root cause. Preserve evidence, contain impact, recover, validate, then investigate why the condition happened and how to prevent/detect recurrence.

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DE16-14 Review / Repair - Terraform and IaC foundations

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What strong evidence shows

Terraform configuration is versioned desired state. fmt/validate checks syntax/style; plan evaluates proposed change; apply changes real infrastructure. Never claim provisioning from a static file.

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